

Bhavdeep Arora

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Education

Toronto Metropolitan University

Toronto, ON

Bachelor of Science in Computer Science, Concentration in Software Engineering

July 2026

- Relevant Coursework: Data Structures & Algorithms, Operating Systems, Databases, Software Engineering, Computer Networks, Distributed Systems, Artificial Intelligence

Experience

Volunteer Data Analyst

May 2026 – Present

Data for Good Toronto

Toronto, ON

- Built a data pipeline to scrape and process ETF data for a nonprofit partner, automating what was previously a manual research workflow
- Cleaned and structured scraped data using Python, handling inconsistent sources to produce a reliable dataset for downstream analysis
- Authored a report summarizing findings and pipeline outputs for the partner organization, translating raw data into actionable insights

Technical Assistant

Sep. 2023 – Dec. 2024

SciXchange

Toronto, ON

- Taught computer science fundamentals, introductory programming, and robotics to students as part of a university STEM outreach initiative
- Designed and delivered hands-on lessons and demos, adapting technical content for a wide range of ages and skill levels
- Reached 500+ students across schools and community programs, contributing to outreach efforts addressing underrepresentation in STEM

Projects

[🔗 bhavdeeparora.dev/projects](https://bhavdeeparora.dev/projects)

Thunderhead | *Go, React, Docker, Redis, Nginx, Typescript* | [🐙 GitHub](#) | [🔗 Live](#)

March 2026 - Present

- Built and open-sourced a reverse proxy for bot detection and mitigation, adopted by 20+ organizations protecting critical internet infrastructure from automated attacks

Groat | *Rust, SQL, Docker, Typescript, Javascript* | [🐙 GitHub](#) | [🔗 Live](#)

June 2026 - Present

- Built a self-hosted proxy that slashes AI API costs for teams running LLM applications at scale, using semantic caching and cost-aware model routing

Funes | *Rust, SQL, Ollama, Typescript, React* | [🐙 GitHub](#) | [🔗 Live](#)

April 2026 - July 2026

- Built a local-first productivity tool that lets you search your entire digital footprint, files, notes, and shell history, in plain English, using vector embeddings with zero cloud dependency

Verrere | *React, TypeScript, PostgreSQL, Prisma* | [🐙 GitHub](#) | [🔗 Live](#)

January 2026 - May 2026

- Built a full-stack app for swipe-based book discovery with a physics-based card deck; grew to 50+ active users syncing reading shelves across devices

Precursor | *Python, Pandas, NumPy, matplotlib* | [🐙 GitHub](#) | [🔗 SSRN](#)

May 2026 – July 2026

- Research project analyzing cross-asset momentum spillover using Granger causality and VAR modeling, confirming 5 commodity-equity relationship hypotheses at $p < 0.05$

Technical Skills

Languages: Python, Go, Java, C#, JavaScript/TypeScript, Bash, Fish, Nim, HTML, CSS/SCSS

DevOps & Tools: Linux, AWS, GCP, Cloudflare, Docker (Swarm, Compose, BuildX), Caddy, Nginx, Traefik, WireGuard, Tailscale, GitHub Actions

Frameworks: .NET, Django, FastAPI, Flask, React, Svelte, Astro, Gin, Celery, Postman, Git, RESTful APIs

Databases: PostgreSQL (PostGIS, PgVector), SQLite, Redis, MySQL, Snowflake